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Patent Public Search | Text View

United States Patent Application Publication

20250274383

Kind Code

A1

Publication Date

August 28, 2025

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NETWORK DISCONNECTION RECOVERY IN AN ACCESS DEVICE

Abstract

An access device communicatively coupled to a local area network (LAN) and to an aggregation device in a service provider network establishes layer 3 connectivity with the service provider network via a main network service. Subsequent to establishing layer 3 connectivity to the service provider network, the access device determines that the main network service has lost layer 3 connectivity to the service provider network. The access device attempts to re-establish layer 3 connectivity with the service provider network using a backup network service.

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Family ID: 1000007739273

Appl. No.: 18/588221

Filed: February 27, 2024

Publication Classification

Int. Cl.: H04L45/28 (20220101); H04L67/51 (20220101)

U.S. Cl.:

CPC H04L45/28 (20130101); H04L67/51 (20220501);

Background/Summary

BACKGROUND

[0001] If an access device, such as cable modem or optical network unit, loses layer 3 connectivity,

a local area network (LAN) associated with the access device also loses layer 3 connectivity and computing devices on the LAN can no longer communicate with computing devices external to the LAN.

SUMMARY

[0002] The examples disclosed herein implement network disconnection recovery in an access device.

[0003] In one implementation a method is provided. The method includes establishing, by an access device communicatively coupled to a local area network (LAN) and to an aggregation device in a service provider network, layer 3 connectivity with the service provider network via a main network service. The method further includes subsequent to establishing layer 3 connectivity to the service provider network, determining, by the access device, that the main network service has lost layer 3 connectivity to the service provider network. The method further includes attempting, by the access device, to re-establish layer 3 connectivity with the service provider network using a backup network service.

[0004] In another implementation an access device is provided. The access device includes a memory, and one or more processor devices operable to establish layer 3 connectivity with a service provider network via a main network service. The one or more processor devices are further operable to, subsequent to establishing layer 3 connectivity to the service provider network, determine that the main network service has lost layer 3 connectivity to the service provider network. The one or more processor devices are further operable to attempt to re-establish layer 3 connectivity with the service provider network using a backup network service.

[0005] In another implementation a non-transitory computer-readable storage medium is provided. The non-transitory computer-readable storage medium includes executable instructions operable to cause one or more processor devices of an access device to establish layer 3 connectivity with a service provider network via a main network service. The instructions are further operable to cause the one or more processor devices to, subsequent to establishing layer 3 connectivity to the service provider network, determine that the main network service has lost layer 3 connectivity to the service provider network. The instructions are further operable to attempt to re-establish layer 3 connectivity with the service provider network using a backup network service.

[0006] Individuals will appreciate the scope of the disclosure and realize additional aspects thereof after reading the following detailed description of the examples in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0008] FIGS. 1A-1C are block diagrams of an environment at different points in time in which network disconnection recovery in an access device can be practiced according to some implementations;

[0009] FIG. 2 is a flowchart of a method for network disconnection recovery in an access device according to one implementation;

[0010] FIG. 3 is a flowchart of a method for network disconnection recovery in an access device according to another implementation; and

[0011] FIG. 4 is a block diagram of an access device suitable for implementing examples according to one implementation.

DETAILED DESCRIPTION

[0012] The examples set forth below represent the information to enable individuals to practice the examples and illustrate the best mode of practicing the examples. Upon reading the following description in light of the accompanying drawing figures, individuals will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0013] Any flowcharts discussed herein are necessarily discussed in some sequence for purposes of illustration, but unless otherwise explicitly indicated, the examples and claims are not limited to any particular sequence or order of steps. The use herein of ordinals in conjunction with an element is solely for distinguishing what might otherwise be similar or identical labels, such as “first message” and “second message,” and does not imply an initial occurrence, a quantity, a priority, a type, an importance, or other attribute, unless otherwise stated herein. The term “about” used herein in conjunction with a numeric value means any value that is within a range of ten percent greater than or ten percent less than the numeric value. As used herein and in the claims, the articles “a” and “an” in reference to an element refers to “one or more” of the element unless otherwise explicitly specified. The word “or” as used herein and in the claims is inclusive unless contextually impossible. As an example, the recitation of A or B means A, or B, or both A and B. The word “data” may be used herein in the singular or plural depending on the context. The use of “and/or” between a phrase A and a phrase B, such as “A and/or B” means A alone, B alone, or A and B together.

[0014] A customer typically connects to an Internet service provider network via a hardware device in the customer premises. The hardware device is often called a customer premises equipment (CPE) or access device. The access device may be, for example, a cable modem or an optical network unit (ONU) or optical network terminal (ONT). The access device may be coupled to a router via a local area network (LAN) port and to an edge device in the service provider's network, such as a cable modem termination system (CMTS) or optical line terminal (OLT) via a wide-area network (WAN) port. Some access devices include router functionality and may be coupled to an access point that communicates with client computing devices.

[0015] A service provider may obtain information from an access device and/or provision or otherwise control the access device using multiple different systems, depending on the particular action that needs to be performed on the access device. In some situations an access device may even require an individual to be able to physically manipulate the access device to cause certain things to happen. In such situations a service provider technician must travel to the access device to perform the desired operation on the access device. This is a relatively expensive and time-consuming process for the service provider, and may result in extended downtime for the customer until the technician arrives and resolves a problem.

[0016] Cloud-based command and control (CaC) technology, such as OpenSync® by way of non-limiting example, available at opensync.io, has been developed that provides cloud-controlled and cloud-managed software to identify and manage connected devices with data security. CaC features may include, by way of non-limiting example, unified device configuration management, device firmware upgrade capabilities, and the like, and eliminates the need to utilize multiple separate systems to control or otherwise configure the access device. Typically, a CaC agent is installed on the access device and the CaC agent communicates with a CaC controller implemented on a remote computing device in the service provider's network, which may sometimes be referred to as cloud CaC software.

[0017] Such CaC software can greatly reduce maintenance and training costs for the service provider, and greatly reduce downtime for a customer. On the access device the CaC agent may be implemented as part of the network services of the access device and is responsible for configuring network interfaces and other network-related operations. Because the CaC agent is part of the network services of the access device, a problem with the CaC agent may impact the ability of the

access device to communicate with the service provider network, which will likewise prevent any computing device on the LAN to which the access device is coupled from reaching the Internet. One such problem may arise when the CaC agent and the CaC controller somehow lose the ability to communicate with one another. This may occur, for example, due to unknown bugs in either the CaC agent or the CaC controller, a fault in the CaC agent or the CaC controller, or the like. [0018] When this happens, the LAN loses Internet access resulting in immediate customer dissatisfaction. Because the issue may be the CaC controller rather than the CaC agent, rebooting the access device or otherwise attempting to diagnose the access device will not resolve the problem. The problem cannot be resolved until the CaC controller, or the connection to the CaC controller, is fixed.

[0019] The examples disclosed herein implement network disconnection recovery in an access device. An access device has two network services, a main network service that implements a cloud CaC software via a CaC agent and a backup network service that lacks the CaC agent and need not communicate with a CaC controller in the cloud, but which does not provide CaC to the access device. During an initialization phase of the access device, the access device establishes layer 3 connectivity with the service provider network via the main network service. The access device continuously monitors the main network service to determine whether layer 3 connectivity has been lost. If so, the access device alters a configuration option to cause the access device to establish layer 3 connectivity using the backup network service during the initialization phase. The access device then causes the access device to re-enter the initialization phase. The access device accesses the configuration option during the initialization phase and determines that the backup network service is to be used rather than the main network service, and establishes layer 3 connectivity using the backup network service. In this manner, when available, the access device may implement CaC functionality that enables the service provider to perform operations on and obtain information from the access device remotely. If the CaC software suffers a problem, the access device can relatively quickly resort to the backup network service to ensure the access device and the LAN to which the access device is coupled has layer 3 connectivity, even in the absence of CaC functionality.

[0020] FIG. 1A is a block diagram on an environment 10 in which network disconnection recovery in an access device can be practiced according to some implementations. The environment 10 includes an access device 12 that is communicatively coupled to a local area network (LAN) 14 and to an aggregation device 16 in a service provider network 17. There may be one or more networks between the access device 12 and the aggregation device 16, such as, by way of non-limiting example, a hybrid-fiber coaxial network, or a distributed access architecture network. The access device 12 includes one or more processor devices 18 (hereinafter referred to as “the processor device 18” for the sake of brevity), one or more memories 20 (hereinafter referred to as “the memory 20” for the sake of brevity), and one or more storage devices 22 (hereinafter referred to as “the storage device 22” for the sake of brevity).

[0021] The access device 12 may comprise, by way of non-limiting example, a cable modem or an optical network unit. In some implementations, the access device 12 may include a router 24 and may connect to the LAN 14 via a connection to an access point 26. The access point 26 may be a wired and/or wireless access point, and may enable connections with a plurality of computing devices 28 via either Wi-Fi, a wired Ethernet cable, or both. In some implementations, the access point 26 may include a router 27. In some implementations, the access point 26 may comprise an OpenSync connected Wi-Fi router. The access device 12 may connect to the access point 26 via a LAN port 30. The LAN port 30 may comprise an Ethernet port. The aggregation device 16 may comprise, by way of non-limiting example, a cable modem termination system (CMTS) or an optical line terminal. Where the aggregation device 16 is a CMTS, the access point 26 may connect to the aggregation device 16 via a WAN port 32, which may comprise, by way of non-limiting example, a coaxial cable port. Where the aggregation device 16 is an optical line terminal, the

wide-area network port **32** may comprise, by way of non-limiting example, an optical fiber port. The access device **12**, among other functionality, provides layer **3** connectivity between the computing devices **28** and computing devices on other networks, such as the Internet **34**. The access device **12** and the aggregation device **16** may communicate via a layer **2** communications protocol.

[0022] Upon power-up, or a reboot, the access device **12** enters an initialization phase that involves loading and initiating an operating system (OS) and otherwise setting up the access device **12** for communications with other devices, such as the access point **26** and the aggregation device **16**. During this initialization phase, the access device **12** initiates either a main network service **36** or a backup network service **38** to establish layer **3** connectivity with the service provider network **17**. In particular, the access device **12** may obtain an IP address from a DHCP server **40** in the service provider network **17** to establish layer **3** connectivity.

[0023] During normal operations, the access device **12** may continuously collect, maintain and send telemetry data to a telemetry service **42** in the service provider network **17**, where such telemetry data may be processed, analyzed, and/or aggregated over time, and presented on a display device for an operator. Each of the telemetry metrics described below may have the same or different collection periods. In some implementations, a default collection time may be, by way of non-limiting example, 30 seconds. Such telemetry data may, by way of non-limiting example, include upstream channel information **44** that includes DOCSIS upstream channel information (e.g., RF level, channel frequency) used by the access device **12** to send data to the aggregation device **16**; downstream channel information **46** (e.g., RF level, channel frequency) used by the access device **12** to receive data from the aggregation device **16**; round-trip (RTT) latency information **48** that identifies round trip latencies between the access device **12** and the computing devices **28**; packet information **50** that identifies the number of transmitted and received packets from each of the computing devices **28**; spectrum information **52** that identifies the RF downstream and upstream spectrum of the access device **12**; aggregation device minimum, maximum, and average roundtrip information **54** that identifies the DOCSIS latency between the access device **12** and the aggregation device **16**; downstream and upstream speed test results; security information of the access device **12**, and the like. Each of the telemetry components **44-54** may comprise processes, such as plug-in modules, that are operable to monitor or otherwise determine the particular telemetry data for which the telemetry component **44-54** is responsible for collecting. The formatting or packaging of the telemetry data on the access device **12** and the streaming of such data to the telemetry service **42** may differ depending on whether the main network service **36** or the backup network service **38** is used to establish layer **3** connectivity to the service provider network **17**.

[0024] During the initialization phase the access device **12** may access a configuration option **56** to determine whether to utilize the main network service **36** or the backup network service **38** to establish layer **3** connectivity. The configuration option **56** may be maintained, for example, on the storage device **22** or stored in a non-volatile memory of the access device **12**. In this example, the configuration option **56** indicates that the access device **12** should utilize the main network service **36** to establish layer **3** connectivity (e.g., "ENABLE_MAIN=Y"). The configuration option **56** also indicates that a main network service (MNS) telemetry plug-in **58** should be used to collect and stream the telemetry data to the telemetry service **42** (e.g., "MNS TEL. PLUG-IN"). If the backup network service **38** were utilized to establish layer **3** connectivity, a backup telemetry plug-in **60** would be used to collect and stream the telemetry data to the telemetry service **42** in lieu of the main network service telemetry plug-in **58**.

[0025] During the device boot up process (i.e., the initialization phase) the access device **12** initiates the main network service **36** to instantiate an executing main network service **62**. The main network service **62** includes a CaC agent **64** that is operable to establish communications with a CaC controller **66** that is executing in the service provider network **17**. The CaC agent **64** receives

configuration information from the CaC controller **66** that allows the CaC agent **64** to configure network interfaces (e.g., bringing interfaces up, creating software bridges), initiate a firewall (e.g., establishing IPtable rules), establish routing rules (e.g., how packets flow between the LAN port **30** and the WAN port **32**), and establish layer **3** connectivity with the service provider network **17**, such as by obtaining an IP address from the DHCP server **40**. If connectivity with the CaC controller **66** is subsequently lost, the configuration information is lost and thus layer **3** connectivity is lost. The CaC agent **64** may attempt to re-establish connectivity to the CaC controller **66**, however, if not successful, layer **3** communications between the access device **12** and the service provider network **17** cannot be established.

[0026] The access device **12** initiates a telemetry collector **68** which in turn instantiates the main network service telemetry plug-in **58** as an executing main network service telemetry process **70**. The access device **12** also initiates a connection monitor **72** that is operable to monitor connectivity and determine if the main network service **62** has lost layer **3** connectivity. The particular monitoring mechanism may differ depending on the particular CaC implementation. In the case of an OpenSync CaC implementation, the connection monitor **72** may issue an OpenvSwitch command, such as an “ovs-vsctl show” command to access the CaC configuration file which contains information indicating whether the CaC connection is established. Other CaC implementations may be monitored based on how such CaC implementations maintain CaC connectivity status.

[0027] While the main network service **62** has layer **3** connectivity, the access device **12** is operable to receive data transmitted by a computing device **28** and forward such data to a destination on another network, such as the Internet **34**, and receive data from another network, such as the Internet **34**, and send such data to a computing device **28**. The telemetry collector **68** periodically, such as every five seconds, 10 seconds, 30 seconds, or any other suitable or desirable period of time, collects and streams the telemetry data to the telemetry service **42**.

[0028] Referring now to FIG. **1B**, for purposes of illustrating network disconnection recovery in the access device **12** according to one implementation, assume that the connection monitor **72** determines that the main network service **62** has lost layer **3** connectivity to the service provider network **17**. This may occur, for example, because the CaC controller **66** has faulted or for some other reason the CaC agent **64** and the CaC controller **66** can no longer communicate with one another. Having lost layer **3** connectivity to the service provider network **17**, the computing devices **28** can no longer communicate with networks that are external to the LAN **14**, such as the service provider network **17** or the Internet **34**, and the telemetry collector **68** can no longer stream the telemetry data to the telemetry service **42**.

[0029] In response to determining that the main network service **62** has lost layer **3** connectivity to the service provider network **17**, the access device **12** alters the configuration option **56** to cause the access device **12** to establish layer **3** connectivity with the service provider network **17** using the backup network service **38** (e.g., “ENABLE_MAIN=N”) that lacks the CaC functionality during the initialization phase. The access device **12** also alters the configuration option **56** to indicate that the backup telemetry plug-in **60** should be used to collect and stream the telemetry data to the telemetry service **42** (e.g., “BACKUP_TEL. PLUG-IN”). The access device **12** then causes the access device **12** to re-enter the initialization phase, such as, by way of non-limiting example, by causing the access device **12** to reboot.

[0030] Referring now to FIG. **1C**, the access device **12** re-enters the initialization phase. The access device **12** determines, based on the configuration option **56**, to establish layer **3** connectivity with the service provider network **17** using the backup network service **38** rather than the main network service **36**. The access device **12** initiates the backup network service **38** to instantiate an executing backup network service **74**. The backup network service **74** lacks the CaC agent. The backup network service **74** configures network interfaces, IPtables, and establishes layer **3** connectivity with the service provider network **17**, such as by obtaining an IP address from the DHCP server **40**.

[0031] The access device **12** also initiates appropriate plug-ins **80-90** to stream, via a backup telemetry process **78**, the telemetry data described above with regard to FIG. **1A**. If the access device **12** also provides router functionality for the LAN **14**, the access device initiates a router **92**. In this manner, the access device **12** automatically recovers from the network disconnection associated with the CaC agent **64** and the CaC controller **66**, without human involvement and with minimal disruption to the LAN **14**.

[0032] The access device **12** also initiates a connection monitor **79** that is operable to monitor connectivity to the CaC cloud system to determine whether the CaC controller **66** (or a subsequently initiated CaC controller in the event the CaC controller **66** terminated), becomes reachable. The particular monitoring mechanism may differ depending on the particular backup network service implementation. In one implementation, the connection monitor **79** may cause a message that requests a reply (e.g., a ping command) to be sent to an IP address associated with the CaC controller **66**. The connection monitor **79** may set a timer. The timer duration value may be randomly selected or may be a fixed value. If the timer expires and a reply has not been received, the connection monitor **79** may determine that the main network service remains unavailable. If a reply is received prior to the timer expiring, the connection monitor **79** determines that cloud connectivity has been re-established and that the main network service may now be available. The access device **12** may then alter the configuration option **56** to cause the access device **12** to establish layer **3** connectivity with the service provider network **17** using the main network service **36** (e.g., "ENABLE_MAIN=Y"). The access device **12** also alters the configuration option **56** to indicate that the MNS telemetry plug-in **58** should be used to collect and stream the telemetry data. The access device **12** then causes the access device **12** to re-enter the initialization phase, such as, by way of non-limiting example, by causing the access device **12** to reboot.

[0033] It is noted that various components, such as the telemetry collector **68**, the connection monitor **72**, the telemetry components **44-54**, and the router **24** are illustrated separately for purposes of illustration. However, because such components are components of the access device **12**, functionality implemented by such components may be attributed to the access device **12** generally. Moreover, in examples where the components comprise software instructions that program the processor device **18** to carry out functionality discussed herein, functionality implemented by the components may be attributed herein to the processor device **18**.

[0034] FIG. **2** is a flowchart of a method for network disconnection recovery in an access device according to one implementation. FIG. **2** will be discussed in conjunction with FIGS. **1A-1C**.

[0035] The access device **12**, communicatively coupled to the LAN **14** and to the aggregation device **16** in the service provider network **17**, establishes layer **3** connectivity with the service provider network **17** via the main network service **36** (FIG. **2**, block **1000**). Subsequent to establishing layer **3** connectivity to the service provider network **17**, the access device **12** determines that the main network service **36** has lost layer **3** connectivity to the service provider network **17** (FIG. **2**, block **1002**). The access device **12** attempts to re-establish layer **3** connectivity with the service provider network **17** using the backup network service **38** (FIG. **2**, block **1004**).

[0036] FIG. **3** is a flowchart of a method for network disconnection recovery in an access device according to another implementation. FIG. **3** will be discussed in conjunction with FIGS. **1A-1C**. The access device **12** begins the initialization phase (FIG. **3**, block **2000**). This may occur, for example, when the access device **12** has been powered on, or causes itself to reboot, or faults and automatically reboots. The initialization phase is prior to a phase when the access device **12** is prepared to communicate data between the LAN **14** and the service provider network **17**. At the beginning of the initialization phase the access device **12** does not have layer **3** connectivity with the service provider network **17**. The access device **12** accesses the configuration option **56** (FIG. **3**, block **2002**). The access device **12** determines, based on the configuration option **56**, that the access device **12** is to enter a CaC mode (e.g., "ENABLE_MAIN=Y") (FIG. **3**, block **2004**).

[0037] The access device **12** initiates the CaC agent **64** (FIG. **3**, block **2006**). The access device **12**

starts the main network service **62** to use the CaC agent **64** (FIG. 3, block **2008**). In this implementation, the access device **12** alters a telemetry configuration option so that the telemetry collector **68** will use the MNS telemetry plug-in **58** (FIG. 3, block **2010**). The access device **12** initiates the connection monitor **72** (FIG. 3, block **2012**). The access device **12** determines whether the CaC agent **64** has connectivity to the CaC controller **66** (FIG. 3, block **2014**). If so, the access device **12** initiates the telemetry collector **68** (FIG. 3, block **2016**). The telemetry collector **68** accesses the telemetry configuration option in the configuration option **56** and determines that the main network service telemetry plug-in **58** is to be used to collect and stream the telemetry data rather than the backup telemetry plug-in **60** (FIG. 3, block **2018**). The telemetry collector **68** instantiates the main network service telemetry plug-in **58** as an executing main network service telemetry process **70** (FIG. 3, block **2020**). The access device **12** initiates the telemetry components **44-54**. The telemetry collector **68** begins streaming the telemetry data via the CaC agent **64** to the telemetry service **42** (FIG. 3, block **2022**).

[0038] If at block **2014** the access device **12** determined that there was no connectivity between the CaC agent **64** and the CaC controller **66**, and thus that layer **3** connectivity is not established, the access device **12** determines if a counter is less than a predetermined threshold (FIG. 3, block **2024**). If so, the access device **12** increments a counter, waits a predetermined amount of time, and attempts to restart the CaC agent (FIG. 3, blocks **2026**, **2028**, **2030**). The process returns to block **2014**. If after a predetermined number of times connectivity between the CaC agent **64** and the CaC controller **66** could not be established, the access device alters the configuration option **56** to cause the access device **12** to establish layer **3** connectivity with the service provider network **17** using the backup network service **38** during the initialization phase (FIG. 3, block **2032**). The access device **12** then causes the access device **12** to reboot (FIG. 3, block **2034**).

[0039] If at block **2014** connectivity between the CaC agent **64** and the CaC controller **66** was established, and subsequent to the blocks **2016-2022**, the access device **12** continuously monitors the connectivity between the CaC agent **64** and the CaC controller **66**. In one implementation, the access device **12** may periodically send to a computing device via the main network service **62**, a message that requests a reply from the computing device. The access device **12** may set a timer. If no reply has been received by the access device **12** prior to the timer expiring, the access device **12** may determine that connectivity between the CaC agent **64** and the CaC controller **66** has been lost.

[0040] The access device **12** may then perform steps similar to blocks **2014**, **2024**, **2026**, **2028** and **2030** in an attempt to determine whether restarting the CaC agent **64** may reestablish connectivity. If not, the access device **12** then performs blocks **2032** and **2034** and reboots. The access device **12** then begins the initialization phase again (FIG. 3, block **2000**). The access device **12** accesses the configuration option **56** (FIG. 3, block **2002**). The access device **12** determines, based on the configuration option **56**, that the access device **12** is to enter the non-CaC mode (e.g., “ENABLE_MAIN=N”) (FIG. 3, block **2004**). The access device **12** starts the backup network service **74** to establish layer **3** connectivity to the service provider network **17** (FIG. 3, block **2036**). The access device **12** alters the telemetry configuration option in the configuration option **56** to use the backup telemetry plug-in **60** (FIG. 3, block **2038**). The access device **12** initiates a connection monitor to determine if there is connectivity to the CaC controller **66** (FIG. 3, blocks **2040**, **2042**). If so, the access device **12** alters the configuration option **56** to cause the access device **12** to establish layer **3** connectivity with the service provider network **17** using the main network service **36** during the initialization phase (FIG. 3, block **2044**). The access device **12** then causes the access device **12** to reboot (FIG. 3, block **2034**).

[0041] If at block **2042** the access device determines that there is no connectivity to the CaC controller **66**, the access device **12** initiates the telemetry collector **76** (FIG. 3, block **2016**). The telemetry collector **76** accesses the telemetry configuration option and determines that the backup telemetry plug-in **60** is to be used to collect and stream the telemetry data rather than the main network service telemetry plug-in **58** (FIG. 3, block **2018**). The telemetry collector **76** instantiates

the backup telemetry plug-in **60** as executing backup telemetry process **78** (FIG. 3, block **2020**). The access device **12** initiates the telemetry components **80-90**. The telemetry collector **76** begins streaming the telemetry data via the backup network service **74** to the telemetry service **42** (FIG. 3, block **2022**).

[0042] At step **2043** the access device **12** may continuously monitor connectivity to the CaC controller **66** to determine if connectivity to the CaC controller **66** becomes available. If at step **2043** the access device **12** determines that connectivity to the CaC controller **66** has been re-established, the access device **12** alters the configuration option **56** to cause the access device **12** to establish layer **3** connectivity with the service provider network **17** using the main network service **36** during the initialization phase (FIG. 3, block **2044**). The access device **12** then causes the access device **12** to reboot (FIG. 3, block **2034**). Otherwise, the telemetry collector **68** continues to stream the telemetry data via the CaC agent **64** to the telemetry service **42** (FIG. 3, block **2022**).

[0043] FIG. 4 is a block diagram of the access device **12** suitable for implementing examples according to one implementation. The access device **12** may comprise, for example, a cable modem or an optical network unit. The access device **12** includes the one or more processor devices **18**, the system memory **20**, and a system bus **94**. The system bus **94** provides an interface for system components including, but not limited to, the system memory **20** and the processor device **18**. The processor device **18** can be any commercially available or proprietary processor.

[0044] The system bus **94** may be any of several types of bus structures that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and/or a local bus using any of a variety of commercially available bus architectures. The system memory **20** may include non-volatile memory **96** (e.g., read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), etc.), and volatile memory **98** (e.g., random-access memory (RAM)). A basic input/output system (BIOS) **100** may be stored in the non-volatile memory **96** and can include the basic routines that help to transfer information between elements within the access device **12**. The volatile memory **98** may also include a high-speed RAM, such as static RAM, for caching data.

[0045] The access device **12** may further include or be coupled to a non-transitory computer-readable storage medium such as the storage device **22**, which may comprise, for example, an internal or external hard disk drive (HDD) (e.g., enhanced integrated drive electronics (EIDE) or serial advanced technology attachment (SATA)), HDD (e.g., EIDE or SATA) for storage, flash memory, or the like. The storage device **22** and other drives associated with computer-readable media and computer-usable media may provide non-volatile storage of data, data structures, computer-executable instructions, and the like.

[0046] A number of modules can be stored in the storage device **22** and in the volatile memory **98**, including an operating system and one or more program modules, such as the connection monitor **72**, which may implement the functionality described herein in whole or in part. All or a portion of the examples may be implemented as a computer program product **102** stored on a transitory or non-transitory computer-usable or computer-readable storage medium, such as the storage device **22**, which includes complex programming instructions, such as complex computer-readable program code, to cause the processor device **18** to carry out the steps described herein. Thus, the computer-readable program code can comprise software instructions for implementing the functionality of the examples described herein when executed on the processor device **18**.

[0047] The access device **12** may also include the LAN port communications interface **30** suitable for communicating with the LAN **14** and the WAN port communications interface **32** suitable for communicating with the aggregation device **16**.

[0048] Individuals will recognize improvements and modifications to the preferred examples of the disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A method, comprising: establishing, by an access device communicatively coupled to a local area network (LAN) and to an aggregation device in a service provider network, layer 3 connectivity with the service provider network via a main network service; subsequent to establishing layer 3 connectivity to the service provider network, determining, by the access device, that the main network service has lost layer 3 connectivity to the service provider network; and attempting, by the access device, to re-establish layer 3 connectivity with the service provider network using a backup network service.
2. The method of claim 1, wherein establishing the layer 3 connectivity with the service provider network via the main network service further comprises establishing the layer 3 connectivity with the service provider network via the main network service during an initialization phase of the access device.
3. The method of claim 1, further comprising: prior to attempting, by the access device, to re-establish layer 3 connectivity with the service provider network using the backup network service, altering, by the access device, a configuration option to cause the access device to establish layer 3 connectivity with the service provider network using the backup network service during an initialization phase; and wherein attempting, by the access device, to re-establish layer 3 connectivity with the service provider network using the backup network service further comprises causing, by the access device, the access device to re-enter the initialization phase.
4. The method of claim 3, wherein causing, by the access device, the access device to re-enter the initialization phase comprises causing, by the access device, the access device to reboot.
5. The method of claim 1, wherein the access device is communicatively coupled to the LAN via a LAN port of the access device and is communicatively coupled to the service provider network via a WAN port of the access device.
6. The method of claim 1, wherein the access device comprises a cable modem or an optical network unit.
7. The method of claim 1, further comprising: accessing, by the access device, telemetry data; and prior to determining, by the access device, that the main network service has lost layer 3 connectivity to the service provider network, sending, by the access device, the telemetry data via the main network service.
8. The method of claim 7, wherein the telemetry data comprises one or more of: an upstream channel frequency used by the access device to send data to the aggregation device, a downstream channel frequency used by the access device to receive data from the aggregation device, round-trip latency times between the access device and one or more wireless computing devices communicatively coupled to the LAN and the access device, packet count information that identifies an aggregate quantity of packets transmitted by a computing device communicatively coupled to the LAN, and a DOCSIS latency that identifies a latency between the access device and the aggregation device.
9. The method of claim 1, wherein subsequently determining, by the access device, that the main network service has lost layer 3 connectivity to the service provider network further comprises accessing, by a command and control agent executing on the access device, configuration data that identifies whether connectivity to a command and control controller in the service provider network exists.
10. The method of claim 1, wherein attempting, by the access device, to re-establish layer 3 connectivity with the service provider network using the backup network service comprises: entering, by the access device, an initialization phase of the access device; accessing a configuration option; determining, based on the configuration option, to establish layer 3 connectivity with the service provider network using the backup network service; and initiating, by

the access device, the backup network service to establish layer 3 connectivity with the service provider network.

11. The method of claim 1, wherein the aggregation device comprises a cable modem termination system or an optical line terminal.

12. The method of claim 1, further comprising: in response to establishing layer 3 connectivity with the service provider network via the main network service, initiating, by the access device, a main network service telemetry process from a plurality of telemetry processes.

13. The method of claim 12, further comprising: subsequent to establishing layer 3 connectivity with the service provider network via the backup network service, initiating, by the access device, a backup telemetry process in lieu of the main network service telemetry process.

14. The method of claim 13, further comprising: subsequent to establishing layer 3 connectivity with the service provider network via the backup network service, continuously monitoring, by the access device, availability of a command and control controller in the service provider network; determining, by the access device, that the command and control controller in the service provider network has become available; and in response to determining, by the access device, that the command and control controller in the service provider network has become available, attempting to re-establish layer 3 connectivity with the service provider network using the main network service.

15. An access device, comprising: a memory; and one or more processor devices operable to: establish layer 3 connectivity with a service provider network via a main network service; subsequent to establishing layer 3 connectivity to the service provider network, determine that the main network service has lost layer 3 connectivity to the service provider network; and attempt to re-establish layer 3 connectivity with the service provider network using a backup network service.

16. The access device of claim 15, wherein the access device comprises a cable modem or an optical network unit.

17. The access device of claim 15, wherein to attempt to re-establish layer 3 connectivity with the service provider network using the backup network service, the one or more processor devices are further operable to: enter an initialization phase of the access device; access a configuration option; determine, based on the configuration option, to establish layer 3 connectivity with the service provider network using the backup network service; and initiate the backup network service to establish layer 3 connectivity with the service provider network.

18. The access device of claim 15, wherein the one or more processor devices are further operable to: in response to establishing layer 3 connectivity with the service provider network via the main network service, initiate a main network service telemetry process from a plurality of telemetry processes.

19. A non-transitory computer-readable storage medium that includes executable instructions operable to cause one or more processor devices of an access device to: establish layer 3 connectivity with a service provider network via a main network service; subsequent to establishing layer 3 connectivity to the service provider network, determine that the main network service has lost layer 3 connectivity to the service provider network; and attempt to re-establish layer 3 connectivity with the service provider network using a backup network service.

20. The non-transitory computer-readable storage medium of claim 19, wherein to attempt to re-establish layer 3 connectivity with the service provider network using the backup network service, the executable instructions are further operable to cause the one or more processor devices to: enter an initialization phase of the access device; access a configuration option; determine, based on the configuration option, to establish layer 3 connectivity with the service provider network using the backup network service; and initiate the backup network service to establish layer 3 connectivity with the service provider network.

21. The non-transitory computer-readable storage medium of claim 19, wherein prior to attempting to re-establish layer 3 connectivity with the service provider network using the backup network

service, the executable instructions are further operable to cause the one or more processor devices to alter a configuration option to cause the access device to establish layer **3** connectivity with the service provider network using the backup network service during an initialization phase; and wherein to attempt to re-establish layer **3** connectivity with the service provider network using the backup network service, the executable instructions are further operable to cause the one or more processor devices to cause the access device to re-enter the initialization phase.
